

CIS 421 Database Management Systems Final Project

Lian Flynn-Mitchell

Dec 10, 2024

Implementation

- **SQL DDL (Data Definition Language) Scripts:** Write SQL scripts to create the database schema, including tables and constraints. Refer to `schema_extended.sql` from Part 1 for guidance.
- **SQL DML (Data Manipulation Language) Scripts:** Populate the database by writing SQL DML scripts. See `data_extended.sql` from Part 1 as a reference.
- **SQL Queries:** Write and test a set of **10 queries**. These should include:
 - Single-table queries
 - Multi-table queries using **JOINS**
 - Queries with **aggregate functions**
 - **Subqueries**
 - **ORDER BY** and **HAVING** clauses

SQL Queries

1. *Single Table* — List All Orders

Retrieving a list of all orders with their associated information in the system.

Query

```
SELECT cstmr_id, order_date, due_date
FROM orders;
```

Output

cstmr_id	order_date	due_date
1	2024-12-01 08:00:00	2024-12-01 10:00:00
2	2024-12-02 09:00:00	2024-12-02 11:00:00
3	2024-12-03 10:00:00	2024-12-03 12:00:00
4	2024-12-04 11:00:00	2024-12-04 13:00:00

2. *Single Table* — List All Customers

Retrieving a list of all customers with their associated information in the system.

Query

```
SELECT customer_id, fname, lname, phone, email
FROM customers;
```

Output

customer_id	fname	lname	phone	email
1	John	Doe	555-1234	john.doe@email.com
2	Jane	Smith	555-5678	jane.smith@email.com
3	Alice	Johnson	555-8765	alice.johnson@email.com
4	Bob	Williams	555-4321	bob.williams@email.com

3. *Multi-Table with JOIN* — View Orders and their Products

Get a list of all orders and their associated products.

Query

```
SELECT orders.order_id, orders.cstmr_id, orders.order_date, products.name,
order_items.quantity
FROM orders
JOIN order_items ON orders.order_id = order_items.ord_id
JOIN products ON order_items.prod_id = products.product_id;
```

Output

order_id	cstmr_id	order_date	name	quantity
1	1	2024-12-01 08:00:00	Sourdough Bread	2
1	1	2024-12-01 08:00:00	Chocolate Chip Muffin	3
2	2	2024-12-02 09:00:00	Whole Wheat Bread	1
2	2	2024-12-02 09:00:00	Pumpkin Spice Muffin	2
3	3	2024-12-03 10:00:00	Baguette	3
3	3	2024-12-03 10:00:00	Banana Nut Muffin	2
4	4	2024-12-04 11:00:00	Sesame Bagel	1
4	4	2024-12-04 11:00:00	Chocolate Croissant	2

4. *Multi-Table with JOIN* — List Orders by Customer

Get a list of all orders with customer names and order dates.

Query

```
SELECT orders.order_id, customers.fname, customers.lname, orders.order_date
FROM orders
JOIN customers ON orders.cstmr_id = customers.customer_id;
```

Output

order_id	fname	lname	order_date
1	John	Doe	2024-12-01 08:00:00
2	Jane	Smith	2024-12-02 09:00:00
3	Alice	Johnson	2024-12-03 10:00:00
4	Bob	Williams	2024-12-04 11:00:00

5. *Aggregate Functions* — Find the Most Expensive Product

Get the average total price of orders for each customer.

Query

```
SELECT name, price
FROM products
WHERE price = (
    SELECT MAX(price)
    FROM products
);
```

Output

name	price
Chocolate Cake	20.00

6. *Aggregate Functions* — Calculate Total Amount per Order

Calculate the total amount of each order.

Query

```
SELECT orders.order_id, SUM(products.price * order_items.quantity) AS
total_amount
FROM order_items
JOIN orders ON order_items.ord_id = orders.order_id
JOIN products ON order_items.prod_id = products.product_id
GROUP BY orders.order_id;
```

Output

order_id	total_amount
1	19.25
2	10.10
3	14.20
4	8.75

7. **Subqueries** — List Customers Who Ordered Products Over a Certain Price

Get a list of customers who ordered any product(s) greater than \$5.

Query

```
SELECT orders.order_id, orders.cstmr_id, orders.order_date, customers.fname,
customers.lname
FROM orders
JOIN customers ON orders.cstmr_id = customers.customer_id
WHERE orders.order_id IN (
    SELECT ord_id
    FROM order_items
    JOIN products ON order_items.prod_id = products.product_id
    WHERE products.price > 5
);
```

Output

order_id	cstmr_id	order_date	fname	lname
1	1	2024-12-01 08:00:00	John	Doe

8. **Subqueries** — List Customers Who Ordered a Specific Product

Get a list of customers who ordered 'Sourdough Bread'.

Query

```
SELECT fname, lname
FROM customers
WHERE customer_id IN (
    SELECT cstmr_id
    FROM orders
    JOIN order_items ON orders.order_id = order_items.ord_id
    JOIN products ON order_items.prod_id = products.product_id
    WHERE products.name = 'Sourdough Bread'
);
```

Output

fname	lname
John	Doe

9. **ORDER BY** and **HAVING** — Find Orders by their Total Quantity

Find orders where the total quantity of items ordered is greater than 2.

Query

```
SELECT orders.order_id, products.name, SUM(order_items.quantity) AS
total_quantity_ordered
FROM order_items
JOIN orders ON order_items.ord_id = orders.order_id
```

```

JOIN products ON order_items.prod_id = products.product_id
GROUP BY orders.order_id, products.name
HAVING total_quantity_ordered > 2
ORDER BY total_quantity_ordered DESC;

```

Output

order_id	name	total_quantity_ordered ▾ 1
3	Baguette	3
1	Chocolate Chip Muffin	3

10. ORDER BY and HAVING — Get Top Ordered Records

List the top 5 most ordered products.

Query

```

SELECT products.name, SUM(order_items.quantity) AS total_quantity_ordered
FROM order_items
JOIN products ON order_items.prod_id = products.product_id
GROUP BY products.name
HAVING total_quantity_ordered > 0
ORDER BY total_quantity_ordered DESC;

```

Output

name	total_quantity_ordered ▾ 1
Chocolate Chip Muffin	3
Baguette	3
Pumpkin Spice Muffin	2
Sourdough Bread	2
Chocolate Croissant	2
Banana Nut Muffin	2
Whole Wheat Bread	1
Sesame Bagel	1